

Education

The University of Texas at Austin (GPA: 3.8/4.0)

May 2027

B.S. Electrical Engineering, B.S. Physics, B.S. Mathematics, Certificate German

Relevant Coursework: Embedded Systems, Solid-State Devices, Power Electronics, Fundamental Electronic Circuits

Graduate Coursework: Magnetic Materials and Devices, Computational Physics, Quantum Mechanics I

Experience

Emerson – Electrical Engineering Design Intern

May 2026–Aug 2026

- Captured a 4–20 mA op-amp and potentiometer signal-conditioning schematic enabling field calibration of set-points of a reverse-engineered raw production PCB
- Troubleshoot and debugged electrical systems using oscilloscopes and logic analyzers to resolve power electronics and control system performance issues
- Collaborated with Production, Manufacturing, and Engineering teams to author product Work Instructions
- Proposed and spearheaded the restructuring of Magtech data organization, improving query speed

Adom Industries Inc. – R&D Design Engineer Intern

May 2025–Aug 2025

- Architected a mixed-signal Vehicle Control Unit (VCU) utilizing an RP5 and RP2040 to bridge low-level CAN bus motor control with high-level ROS path planning
- Designed and routed custom mixed-signal PCBs for LiDAR and motor control, implementing digital isolation to successfully mitigate electromagnetic interference (EMI) in a 48V power system

Sandia National Laboratories – R&D Computational Engineer

May 2024–Aug 2024

- Built a fast current-pulse generator modeling capacitor discharges, synthesizing 10k+ waveforms for ML surrogate modeling and Uncertainty Quantification
- Analyzed fluid RTIs and RMIIs to diagnose simulation inaccuracies

Projects and Organizations

Safety-Critical UAV Flight Controller

- Architected deterministic flight control firmware in Rust utilizing RTIC (Real-Time Interrupt-driven Concurrency); applied Rate Monotonic Scheduling (RMS) to safely isolate safety-critical attitude loops from ML inference tasks
- Designing an ML-integrated flight controller with a nonlinear Kalman filter fusing noisy IMU measurements
- Minimizing ESC-induced EMI on a custom 4-layer PCB with dedicated ground planes and digital isolators

Embedded Racing Simulator

- Implemented multi-node CAN bus network, debugging bus arbitration and extending the data payload to support higher-bandwidth real-time telemetry between the steering wheel, pedal box, and central compute
- Designed custom PCBs with SPI in KiCAD for steering wheel-mounted TM4Cs, integrating CAN transceivers and signal conditioning for a rotary encoder-based throttle/brake input

High-Power E-Bike Boost Converter

- Engineered a custom DC-DC boost converter PCB, suppressing radiated EMI by minimizing the high di/dt critical loop and gate-drive parasitic inductance

Banerjee Lab – 2D Material Synthesis & Characterization

Sep 2025–Present

- Characterized and grew MoS₂ monolayers via CVD to optimize carrier type, doping, and bandgap
- Employed XRD, Raman, and Photoluminescence spectroscopy for material analysis

FSAE Longhorn Racing – Dynamics + Vehicle Modeling Engineer, Steering Lead

Sep 2022–Aug 2025

- Owned, designed, and manufactured the steering system from steel to a carbon-fiber layup with a metal 3D-printed bevel gearbox, cutting weight 68% at equal torsional rigidity

Skills

Languages: C, C++, ARM, Rust, Python, MATLAB, Bash, JS, HTML, CSS, LaTeX

Tools: KiCAD, Cadence, Git/GitHub, LTSpice, Oscilloscopes, Logic Analyzers, Multimeters, Soldering, Power Supplies